Lab 8: 2D Geometric Transformations

Computer Graphics Lab, Wolfram Mathematica

In[1]:=
```
SetOptions[EvaluationNotebook[], ShowCellLabel → True];
```

In[2]:=
```
SetOptions[{Plot, Plot3D, ContourPlot, ContourPlot3D, DensityPlot,
   ParametricPlot, ParametricPlot3D, ListPlot, ListLinePlot, ListPointPlot3D,
   Graphics, Graphics3D, PolarPlot}, ImageSize → Small, AxesLabel → {"x", "y", "z"}];
```

In[3]:=
```
(* Problem 1 Rotate an equilateral triangle RT=
 2 at the distance RR=7/2 from the origin. Rotate the triangle around itself.
    The code highlighted by Cyan is attached to the Lab manual.
*)
```

In[4]:=
```
RT1 = 2; RR = 7 / 2; Rad1 = {RR, 0};
```

In[5]:=
```
(* Center *)
```

In[6]:=
```
CentRot1 = {0, 0};
```

In[7]:=
```
OTri1 = RegularPolygon[CentRot1, RT1, 3]
```

Out[7]=
```
RegularPolygon[{0, 0}, 2, 3]
```

In[8]:=
```
(* Center point  *)
```

In[9]:=
```
OCent1 = {Red, Point[CentRot1]}
```

Out[9]=
```
{■, Point[{0, 0}]}
```

In[10]:=
```
(* The Triangle and the center point *)
```

In[11]:=
```
O1 = {OTri1, OCent1}
```

Out[11]=
```
{RegularPolygon[{0, 0}, 2, 3], {■, Point[{0, 0}]}}
```

In[12]:=
```
(* ATTN: Compose the rotation function *)
```

In[13]:=
```
ORot1[a_] := GeometricTransformation[O1,
  Composition[RotationTransform[a], TranslationTransform[Rad1]]]
```

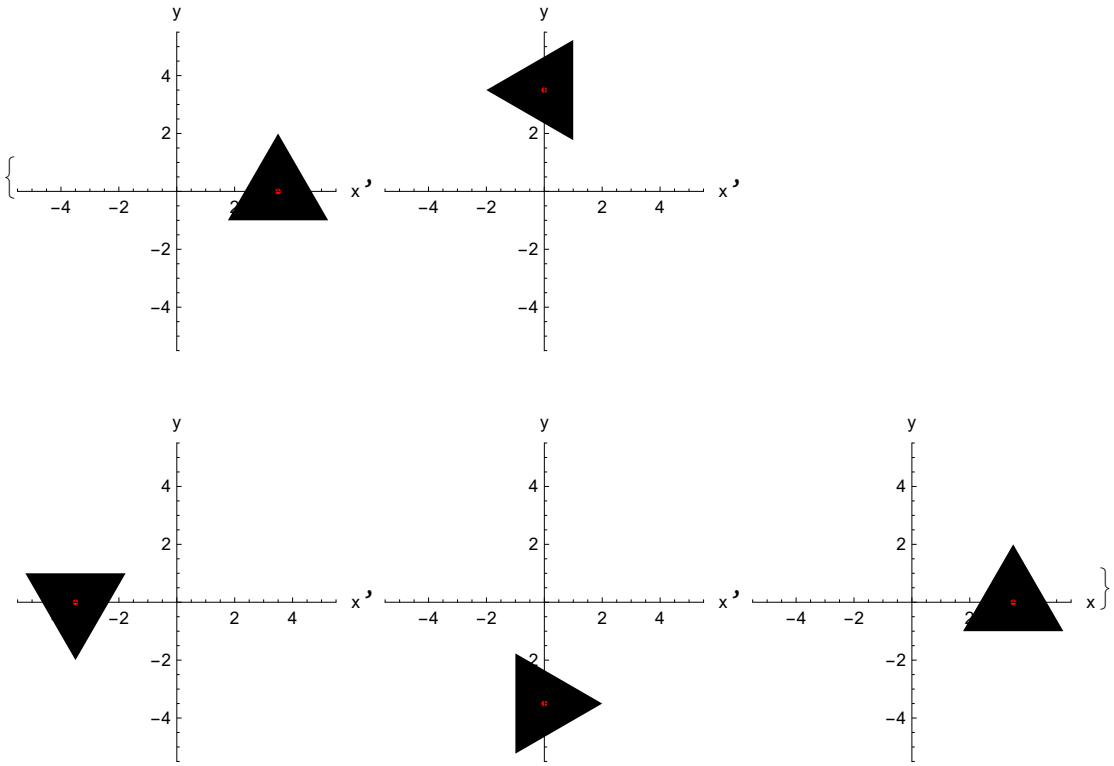
In[14]:=
```
(* Define the graphics function  *)
```

In[15]:=
```
GTri1[a_] := Graphics[ORot1[a], Axes → True, AxesOrigin → {0, 0}, PlotRange → 5.5]
```

In[16]:=
```
(* Check  *)
```

In[17]:=
```
Table[GTri1[a], {a, 0, 2 Pi, Pi / 2}]
```

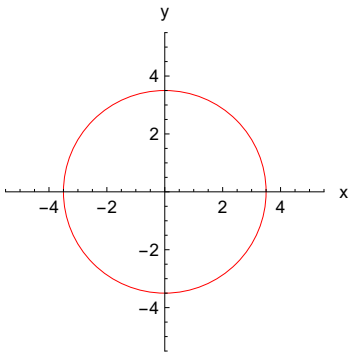
Out[17]=



In[18]:=
```
OCir1 := {Red, Circle[CentRot1, RR]}
```

In[19]:=
```
GCir1 = Graphics[OCir1, Axes → True, AxesOrigin → {0, 0}, PlotRange → 5.5]
```

Out[19]=

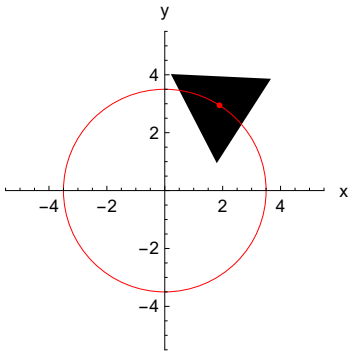


In[20]:=
```
(* Animate triangle moving along the circular
 trajectory with the radius 7/2. Display the Circle[RR[[1]]]  *)
```

In[21]:=
```
OT1[t_] := Graphics[{ORot1[t], OCir1}, Axes → True, AxesOrigin → {0, 0}, PlotRange → 5.5]
```

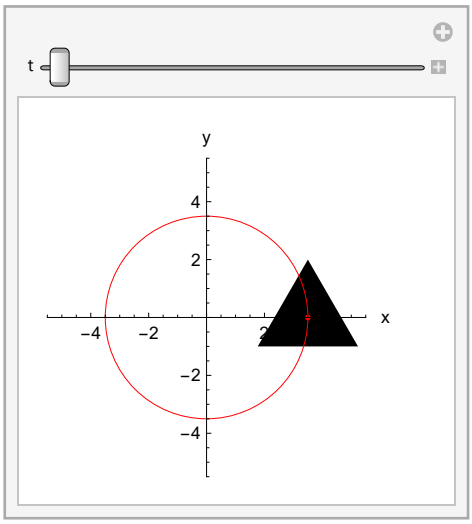
In[22]:=
```
OT1[1]
```

Out[22]=



In[23]:=
```
Manipulate[OT1[t], {t, 0, 2 Pi, 0.1}]
```

Out[23]=



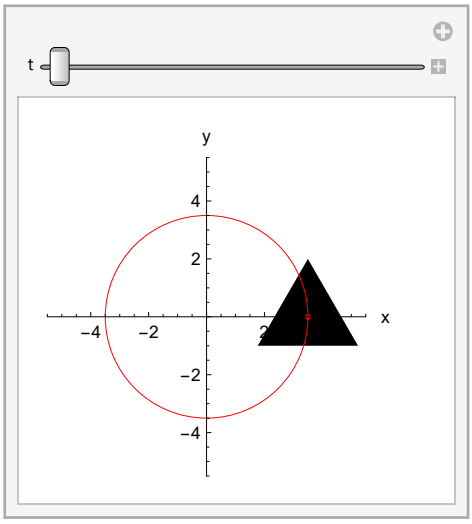
In[24]:=
```
ORot2[a_] := GeometricTransformation[O1,
  Composition[RotationTransform[a], TranslationTransform[Rad1], RotationTransform[a]]]
```

In[25]:=
```
GTri2[a_] := Graphics[ORot2[a], Axes → True, AxesOrigin → {0, 0}, PlotRange → 5.5]
```

In[26]:=
```
OT2[t_] := Graphics[{ORot2[t], OCir1}, Axes → True, AxesOrigin → {0, 0}, PlotRange → 5.5]
```

In[27]:=
```
Manipulate[OT2[t], {t, 0, 2 Pi, 0.1}]
```

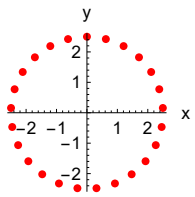
Out[27]=



In[28]:=
```
(* Problem 2 *)
```

In[29]:=
```
Graphics[{Red, PointSize[0.05], Point[CirclePoints[2.5, 25]]},
 ImageSize → Tiny, Axes → True]
```

Out[29]=

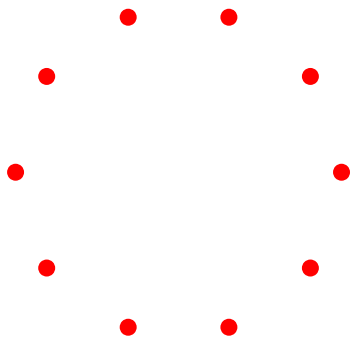


In[30]:=
```
Points = {Red, PointSize[0.05], Point[CirclePoints[2.5, 10]]}
```

Out[30]=
```
{■, PointSize[0.05],
 Point[{{0.772542, -2.37764}, {2.02254, -1.46946}, {2.5, 0.}, {2.02254, 1.46946},
   {0.772542, 2.37764}, {-0.772542, 2.37764}, {-2.02254, 1.46946},
   {-2.5, 3.06162 × 10^-16}, {-2.02254, -1.46946}, {-0.772542, -2.37764}}]}
```

In[31]:=
```
Graphics[Points]
```

Out[31]=

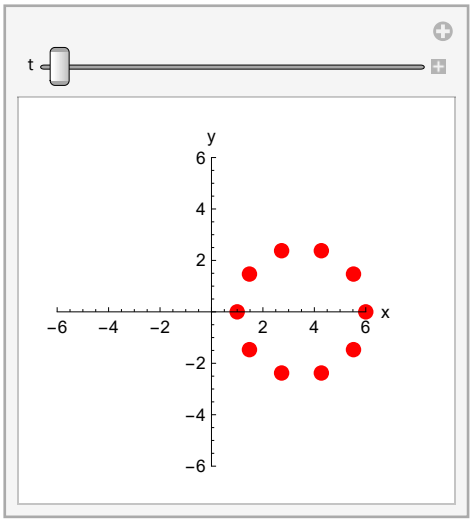


In[32]:=
```
RotPoints[a_] := GeometricTransformation[Points,
  Composition[RotationTransform[a], TranslationTransform[Rad1], RotationTransform[a]]]
```

In[33]:=
```
GPoints[a_] := Graphics[RotPoints[a], Axes → True, AxesOrigin → {0, 0}, PlotRange → 6]
```

In[34]:=
```
Manipulate[GPoints[t], {t, 0, 2 Pi, 0.1}]
```

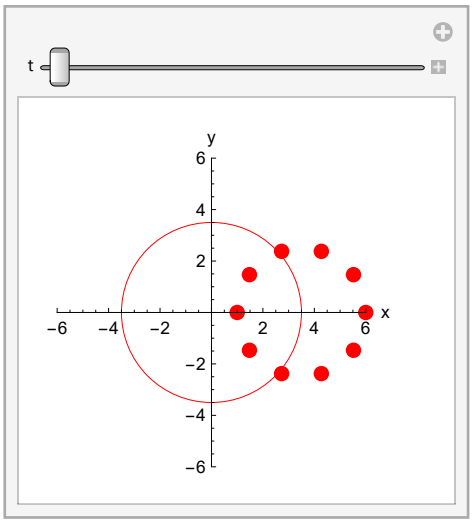
Out[34]=



In[35]:=
```
OTPoints[t_] :=
 Graphics[{RotPoints[t], OCir1}, Axes → True, AxesOrigin → {0, 0}, PlotRange → 6]
```

In[36]:=
```
Manipulate[OTPoints[t], {t, 0, 2 Pi, 0.1}]
```

Out[36]=



In[37]:=
```
P1 = 1.0 * CirclePoints[RT1, 10]
```

Out[37]=
```
{{0.618034, -1.90211}, {1.61803, -1.17557}, {2., 0.},
 {1.61803, 1.17557}, {0.618034, 1.90211}, {-0.618034, 1.90211},
 {-1.61803, 1.17557}, {-2., 0.}, {-1.61803, -1.17557}, {-0.618034, -1.90211}}
```

In[38]:=
```
nCP1 = Dimensions[P1][[1]]
```

Out[38]=
```
10
```

In[39]:=
```
PointsC1 = {PointSize[0.08], Point[P1, VertexColors → RandomColor[nCP1]]}
```

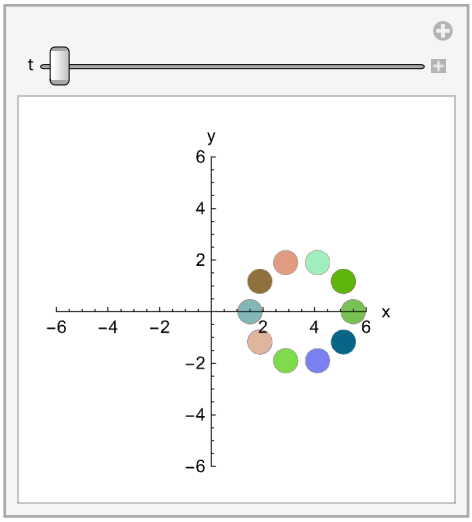
Out[39]=
```
{PointSize[0.08],
 Point[{{0.618034, -1.90211}, {1.61803, -1.17557}, {2., 0.}, {1.61803, 1.17557},
   {0.618034, 1.90211}, {-0.618034, 1.90211}, {-1.61803, 1.17557},
   {-2., 0.}, {-1.61803, -1.17557}, {-0.618034, -1.90211}},
  VertexColors → {■, ■, ■, ■, ■, ■, ■, ■, ■, ■}]}
```

In[40]:=
```
RotPointsC1[a_] := GeometricTransformation[PointsC1,
  Composition[RotationTransform[a], TranslationTransform[Rad1], RotationTransform[a]]]
```

In[41]:=
```
GPoints1[a_] := Graphics[RotPointsC1[a], Axes → True, AxesOrigin → {0, 0}, PlotRange → 6]
```

In[42]:=
```
Manipulate[GPoints1[t], {t, 0, 2 Pi, 0.1}]
```

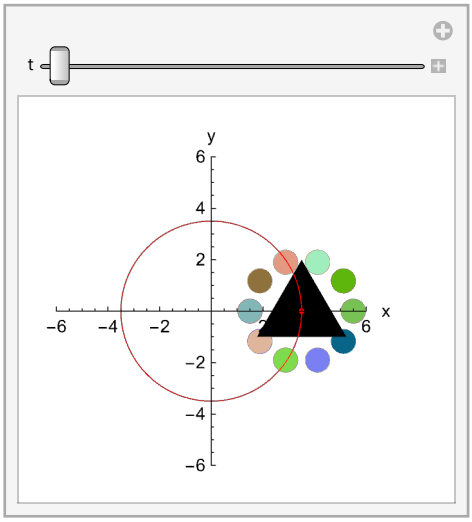
Out[42]=



In[43]:=
```
GAll[t_] := Show[{GPoints1[t], OT1[t]}]
```

In[44]:=
```
Manipulate[GAll[t], {t, 0, 2 Pi, 0.1}]
```

Out[44]=



In[45]:=
```
Rot1[O_, R_, a_] := Module[{ORot, GrO, a1},
  ORot[a1_] := GeometricTransformation[O,
    Composition[RotationTransform[a], TranslationTransform[{R, 0}]]];
  GrO[a1_] := Graphics[ORot[a1], Axes → True, AxesOrigin → {0, 0}, PlotRange → 5.5];
  GrO[a]]
```

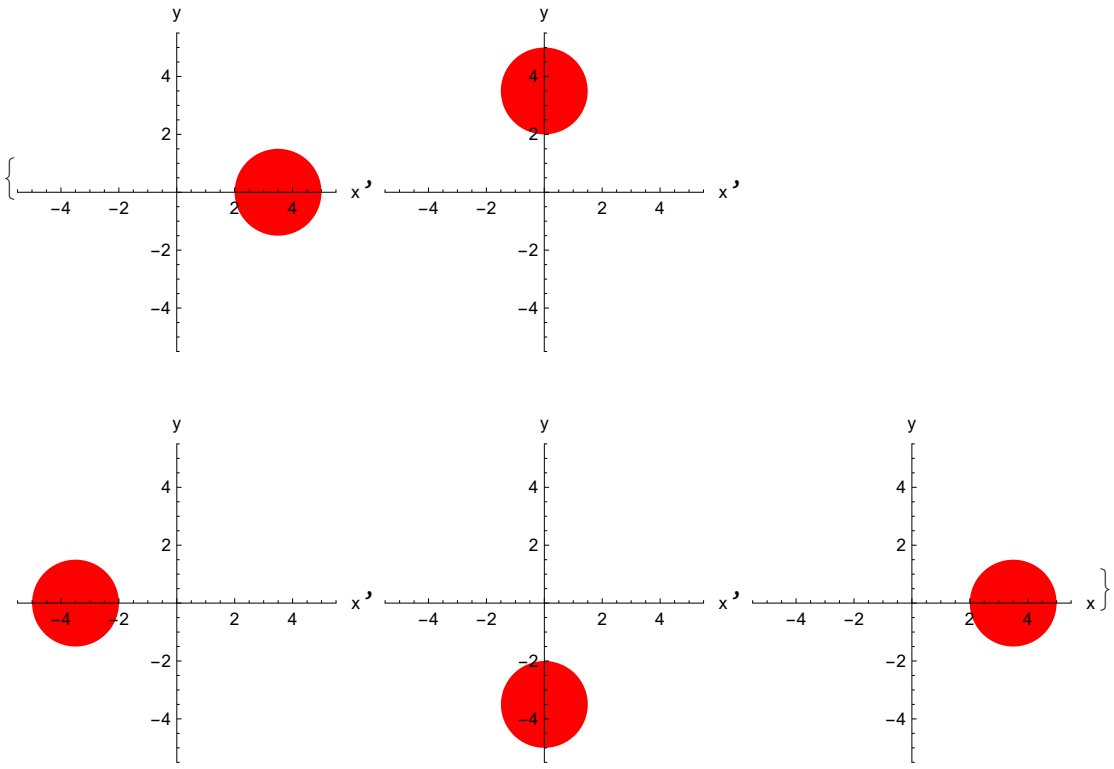
In[46]:=
```
Rot1F[O_, R_] := Module[{ORot, GrO, a1},
  ORot[a1_] := GeometricTransformation[O,
    Composition[RotationTransform[a1], TranslationTransform[{R, 0}]]];
  GrO[a1] := Graphics[ORot[a1], Axes → True, AxesOrigin → {0, 0}, PlotRange → 5.5];
  GrO]
```

In[47]:=
```
Disk1 = {Red, Disk[{0, 0}, 1.5]}
```

Out[47]=
```
{■, Disk[{0, 0}, 1.5]}
```

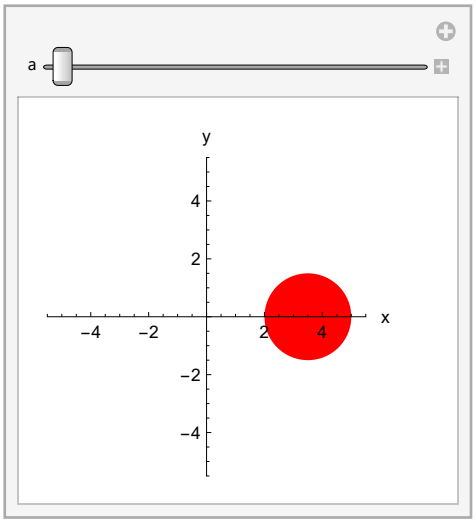
In[48]:=
```
Table[Rot1[Disk1, RR, a], {a, 0, 2 Pi, Pi / 2}]
```

Out[48]=



In[49]:=
```
Manipulate[Rot1[Disk1, RR, a], {a, 0, 2 Pi, Pi / 4}]
```

Out[49]=

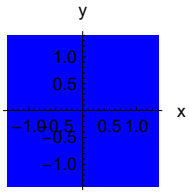


In[50]:=
```
(*Problem 3.1*)
```

In[51]:=
```
O3 = {Blue, RegularPolygon[2, 4]};
```

In[52]:=
```
Graphics[O3, ImageSize → Tiny, Axes → True]
```

Out[52]=



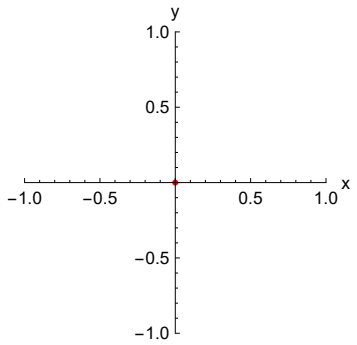
In[53]:=
```
CenterRot = {5, 5}; RD = {4, 0};
```

In[54]:=
```
Dist = CenterRot + RD;
```

In[55]:=
```
O3Zero = {Red, PointSize[0.02], Point[{0, 0}]};
```

In[56]:=
```
Graphics[O3Zero, Axes → True]
```

Out[56]=

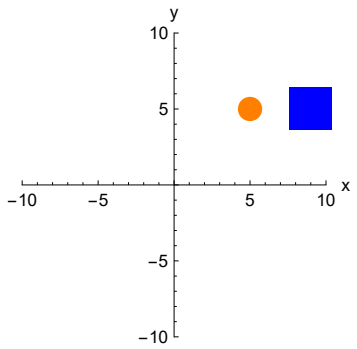


In[57]:=
```
O3Trans = GeometricTransformation[O3, TranslationTransform[Dist]];
```

In[58]:=
```
OCenter3 = {Orange, PointSize[0.08], Point[CenterRot]};
```

In[59]:=
```
Graphics[{O3Trans, OCenter3}, Axes → True, PlotRange → 10]
```

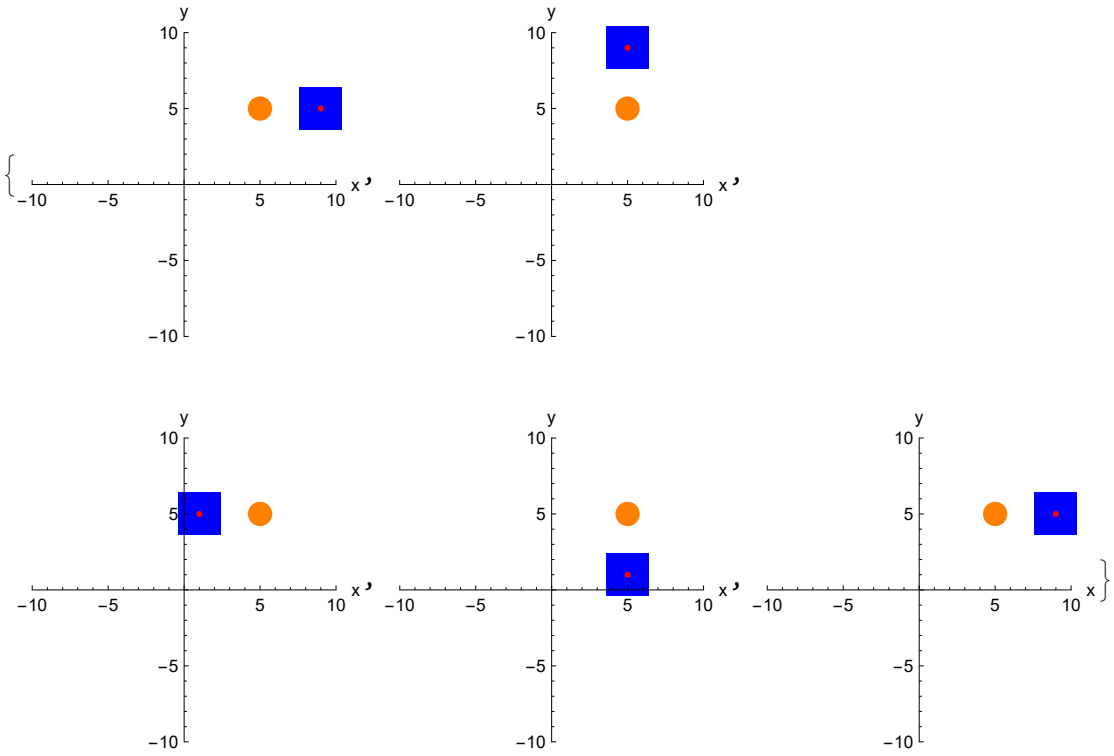
Out[59]=



In[60]:=
```
ORot3[a_] := GeometricTransformation[{O3, O3Zero},
   Composition[RotationTransform[a, CenterRot], TranslationTransform[Dist]]];
```

In[61]:=
```
Table[Graphics[{ORot3[a], OCenter3}, Axes → True, PlotRange → 10], {a, 0, 2 Pi, Pi / 2}]
```

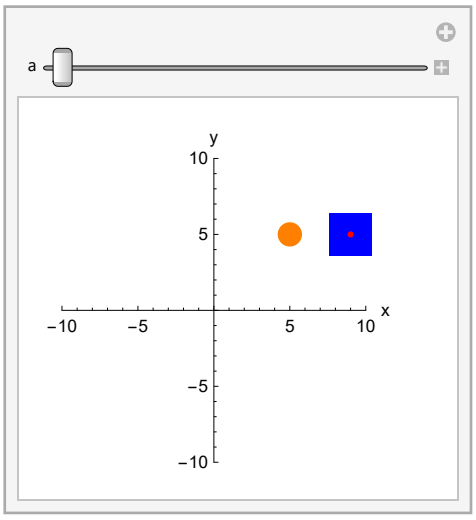
Out[61]=



In[62]:=
```
GORot3[a_] := Graphics[{ORot3[a], OCenter3}, Axes → True, PlotRange → 10]
```

In[63]:=
```
Manipulate[GORot3[a], {a, 0, 2 Pi}]
```

Out[63]=

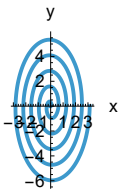


In[64]:=
```
(* Problem 3.2 *)
```

In[65]:=
```
T3[t_] := 0.1 {t * Cos[t], 2 * t * Sin[t]}
```

In[66]:= **PT3 = ParametricPlot[T3[t], {t, 0, 10 Pi}, ImageSize → Tiny]**

Out[66]=



In[67]:= **OT3 = PT3[[1]]**

Out[67]=

```
{{{{}, {}, {Directive[Opacity[1.], AbsoluteThickness[2], FaceForm[Opacity[0.3]], ■],
    Line[{{0., 0.}, {0.000963474, 0.0000185671}, {0.00192668, 0.0000742649}, {0.00288935, 0.000167083},
      {0.00385121, 0.000297004}, {0.004812, 0.000464005}, {0.00577145, 0.000668053}, ··· 2002 ··· ,
      {3.12856, -0.405367}, {3.13164, -0.337994}, {3.13436, -0.270535}, {3.13672, -0.202999},
      {3.13871, -0.135392}, {3.14033, -0.0677232}, {3.14159, -7.69468×10^-15}}]}}}, {}}
```

Size in memory: 39.6 kB | + Show more | Show all | Iconize | Store full expression in notebook

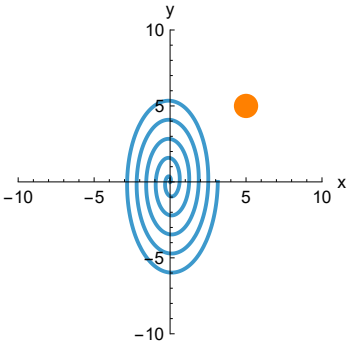
In[68]:= **Graphics[OT3, ImageSize → Tiny]**

Out[68]=



In[69]:= **Graphics[{OT3, OCenter3}, Axes → True, PlotRange → 10]**

Out[69]=



In[70]:= **OT3**

Out[70]=

```
{{{{}, {}, {Directive[Opacity[1.], AbsoluteThickness[2], FaceForm[Opacity[0.3]], ■],
    Line[{{0., 0.}, {0.000963474, 0.0000185671}, {0.00192668, 0.0000742649}, {0.00288935, 0.000167083},
      {0.00385121, 0.000297004}, {0.004812, 0.000464005}, {0.00577145, 0.000668053}, ··· 2002 ··· ,
      {3.12856, -0.405367}, {3.13164, -0.337994}, {3.13436, -0.270535}, {3.13672, -0.202999},
      {3.13871, -0.135392}, {3.14033, -0.0677232}, {3.14159, -7.69468×10^-15}}]}}}, {}}
```

Size in memory: 39.6 kB | + Show more | Show all | Iconize | Store full expression in notebook

In[71]:=
```
OCenter3
```

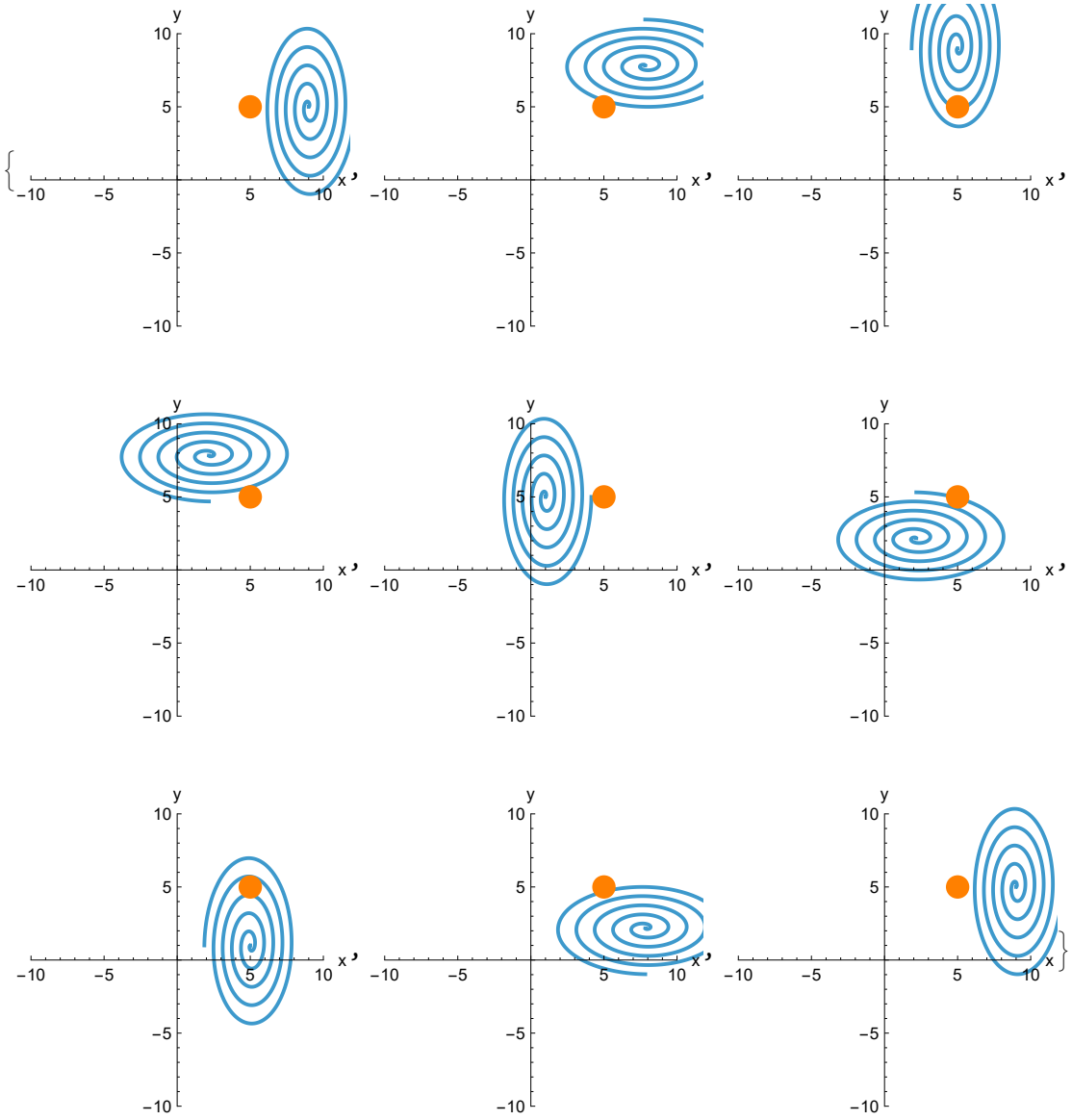
Out[71]=

```
{■, PointSize[0.08], Point[{5, 5}]}
```

In[72]:=
```
ORot4[a_] := GeometricTransformation[OT3, Composition[
    RotationTransform[a, CenterRot], TranslationTransform[Dist], RotationTransform[a]]]
```

In[73]:=
```
Table[Graphics[{ORot4[a], OCenter3}, Axes → True, PlotRange → 10], {a, 0, 2 Pi, Pi / 4}]
```

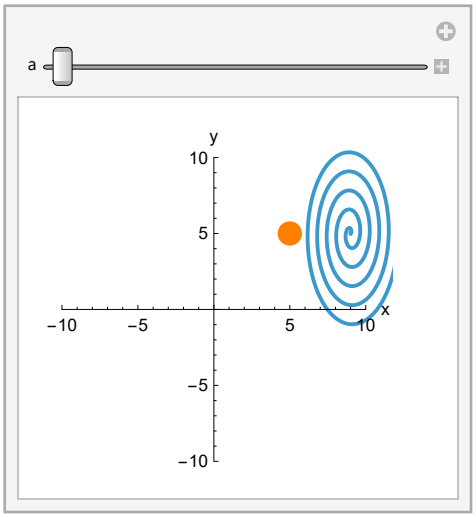
Out[73]=



In[74]:=
```
G3[a_] := Graphics[{ORot4[a], OCenter3}, Axes → True, PlotRange → 10]
```

In[75]:=
```
Manipulate[G3[a], {a, 0, 2 Pi}]
```

Out[75]=



In[76]:=
```
(* Problem 4 *)
```

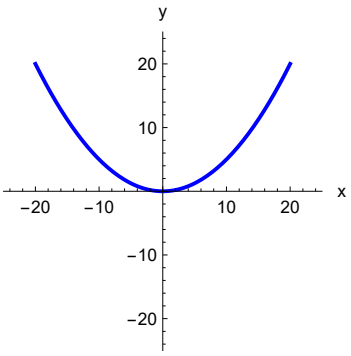
In[77]:=
```
Clear[Ob]
```

In[78]:=
```
Ob[t_] := 4 {Cos[2 t], Sin[4 t]}
```

In[79]:=
```
Traj[t_] := {t, 0.05 t^2}
```

In[80]:=
```
Tra = ParametricPlot[Traj[t], {t, -20, 20}, PlotRange → 25, PlotStyle → {Blue, Thick}]
```

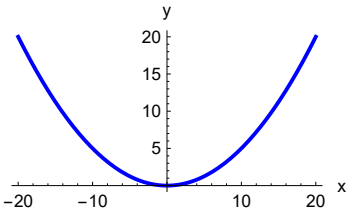
Out[80]=



In[81]:=
```
OTra = Tra[[1]];
```

In[82]:=
```
Graphics[OTra, Axes → True]
```

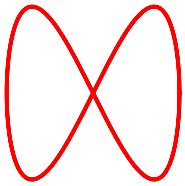
Out[82]=



In[83]:=
```
Curve[t_] := {4 Cos[2 t], 4 Sin[4 t]}
```

In[84]:=
```
P = ParametricPlot[Curve[t], {t, 0, 2 Pi}, PlotStyle → Red, Axes → False, ImageSize → Tiny]
```

Out[84]=



In[85]:=
```
a = Pi / 2
```

Out[85]=

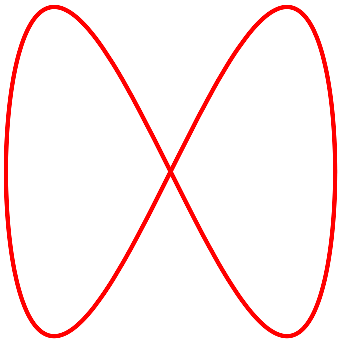
$$\frac{\pi}{2}$$

In[86]:=
```
MTran[t_, a_] := Composition[TranslationTransform[Traj[t]], RotationTransform[a]]
```

In[87]:=
```
Ob = P[[1]];
```

In[88]:=
```
Graphics[Ob]
```

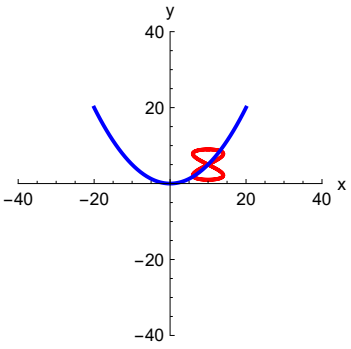
Out[88]=



In[89]:=
```
Trans[t_, a_] := GeometricTransformation[Ob, MTran[t, a]]
```

In[90]:=
```
G1 = Graphics[{Trans[10, Pi / 2], OTra}, Axes → True, PlotRange → 40]
```

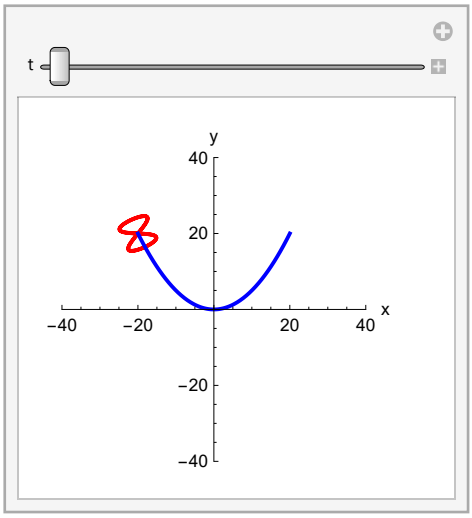
Out[90]=



In[91]:=
```
GTransAll[t_] := Graphics[{Trans[t, t], OTra}, Axes → True, PlotRange → 40]
```

In[92]:=
```
Manipulate[GTransAll[t], {t, -20, 20}]
```

Out[92]=

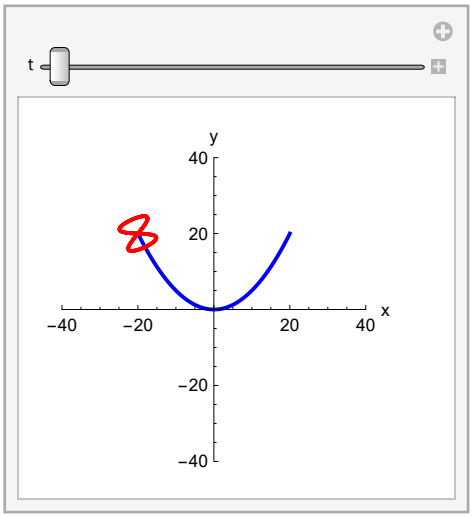


In[93]:=
```
TransNew[t_, a_, b_] := GeometricTransformation[{OTra, Trans[t, a]}, RotationTransform[b]]
```

In[94]:=
```
GTransAllNew[t_] := Graphics[TransNew[t, t, 20 + t], Axes → True, PlotRange → 40]
```

In[95]:=
```
Manipulate[GTransAllNew[t], {t, -20, 20}]
```

Out[95]=



In[96]:=
```
(* Problem 5 *)
```

In[97]:=
```
Rot2[ONew_, OTNew_, Traj_, t_, a_] := Module[{MTran, t1, a1, Trans, G1},
  MTran[t1_, a1_] := Composition[TranslationTransform[Traj[t1]], RotationTransform[a1]];
  Trans[t1_, a1_] := GeometricTransformation[ONew, MTran[t1, a1]];
  G1[t1_, a1_] := Graphics[{Trans[t1, a1], OTNew}, Axes → True, PlotRange → 60];
  G1]
```

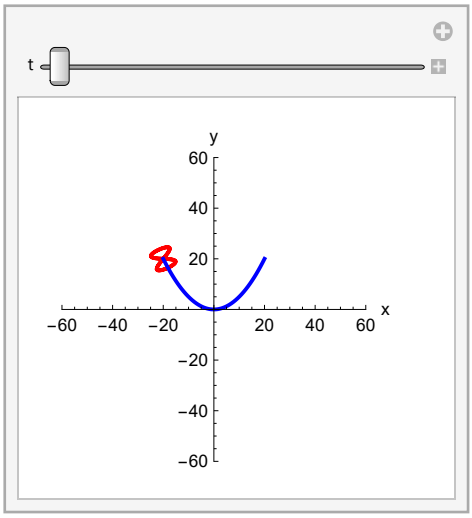
In[98]:=
```
M2 = Rot2[Ob, OTra, Traj, t, a]
```

Out[98]=

```
G1$14351
```

In[99]:=
```
Manipulate[M2[t, t], {t, -20, 20}]
```

Out[99]=



In[100]:=
```
(* Problem 6.1 *)
```

In[101]:=
```
a = 40
b = 0.5
```

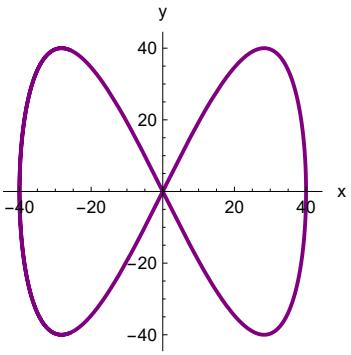
Out[101]=
```
40
```

Out[102]=
```
0.5
```

In[103]:=
```
TrajNew[t_] := a {Cos[2 t], Sin[4 t]}
ObNew[s_] := b {s, 0.05 s^2}
```

In[105]:=
```
Try1 = ParametricPlot[TrajNew[t], {t, -2, 2}, PlotStyle → Purple]
```

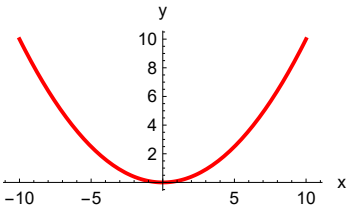
Out[105]=



In[106]:=

```
Try2 = ParametricPlot[ObNew[t], {t, -20, 20}, PlotStyle → Red]
```

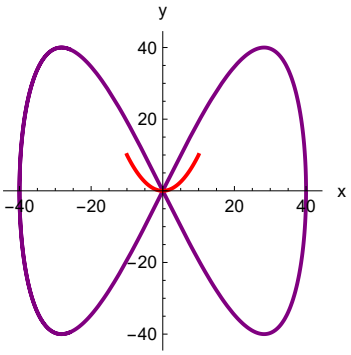
Out[106]=



In[107]:=

```
Show[{Try1, Try2}]
```

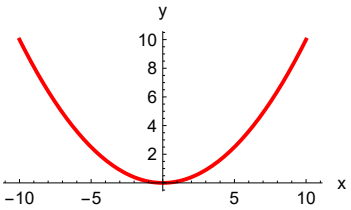
Out[107]=



In[108]:=

```
pp61 = ParametricPlot[ObNew[t], {t, -20, 20}, PlotStyle → Red]
```

Out[108]=



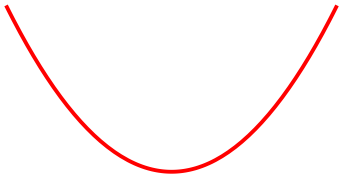
In[109]:=

```
o61 = pp61[[1]];
```

In[110]:=

```
Graphics[o61]
```

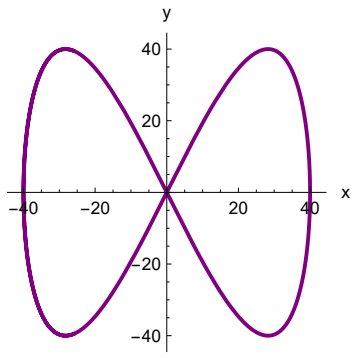
Out[110]=



In[111]:=

```
pp62 = ParametricPlot[TrajNew[t], {t, -2, 2}, PlotStyle → Purple]
```

Out[111]=



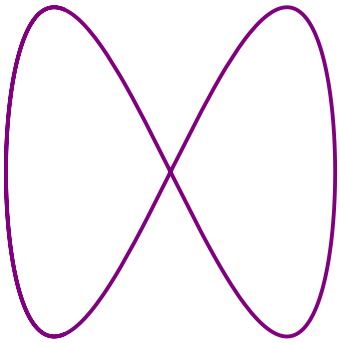
In[112]:=

```
o62 = pp62[[1]];
```

In[113]:=

```
Graphics[o62]
```

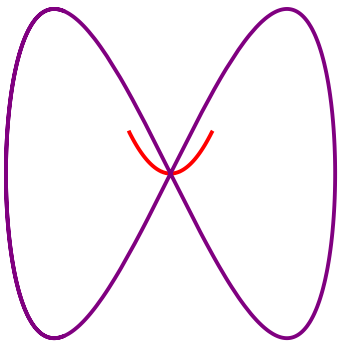
Out[113]=



In[114]:=

```
Graphics[{o61, o62}]
```

Out[114]=



In[115]:=

```
M6 = Rot2[o61, o62, TrajNew, t, a]
```

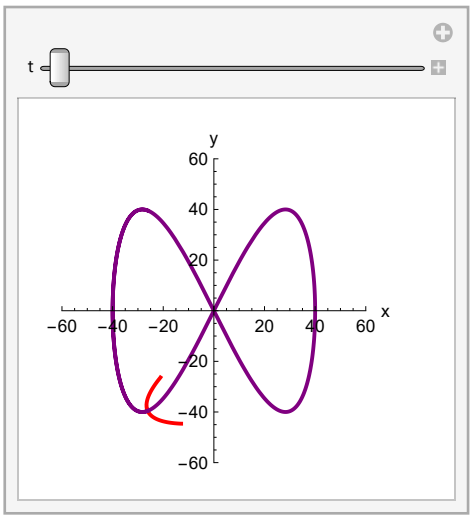
Out[115]=

```
G1$15032
```

In[116]:=

```
Manipulate[M6[t / 10, t], {t, -20, 20}]
```

Out[116]=



In[117]:=

```
(* Problem 7 - Bonus *)
```

In[118]:=

```
Rot5[O1_, O2_, OTraj_, Traj_, trange_] :=
 Module[{TAll, MTran1, MTran2, Trans1, Trans2, t1, a1, GAll},
  MTran1[t1_, a1_] :=
   Composition[TranslationTransform[Traj[-t1 + trange]], RotationTransform[a1]];
  Trans1[t1_, a1_] := GeometricTransformation[O1, MTran1[t1, a1]];

  MTran2[t1_, a1_] :=
   Composition[TranslationTransform[Traj[t1 - trange]], RotationTransform[a1]];
  Trans2[t1_, a1_] := GeometricTransformation[O2, MTran2[t1, a1]];

  GAll[t1_, a1_] :=
   Graphics[{Trans1[t1, a1], Trans2[t1, a1], OTraj}, Axes → True, PlotRange → 60];

  GAll]
```

In[119]:=

```
pp71 = {Red, RegularPolygon[{0, 0}, 20, 4]}
```

Out[119]=

```
{■, RegularPolygon[{0, 0}, 20, 4]}
```

In[120]:=

```
pp72 = {Blue, RegularPolygon[{0, 0}, 20, 6]}
```

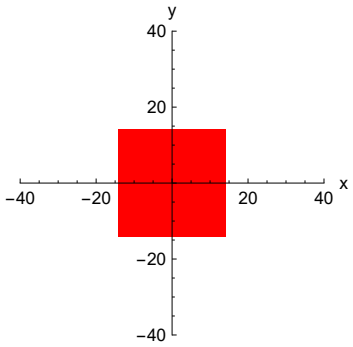
Out[120]=

```
{■, RegularPolygon[{0, 0}, 20, 6]}
```

In[121]:=

```
Graphics[pp71, PlotRange → 40, Axes → True]
```

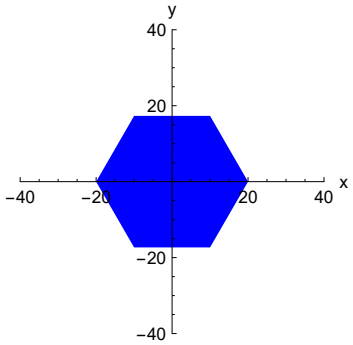
Out[121]=



In[122]:=

```
Graphics[pp72, PlotRange → 40, Axes → True]
```

Out[122]=



In[123]:=

```
trange = 20
```

Out[123]=

```
20
```

In[124]:=

```
M7 = Rot5[pp71, pp72, o62, TrajNew, trange]
```

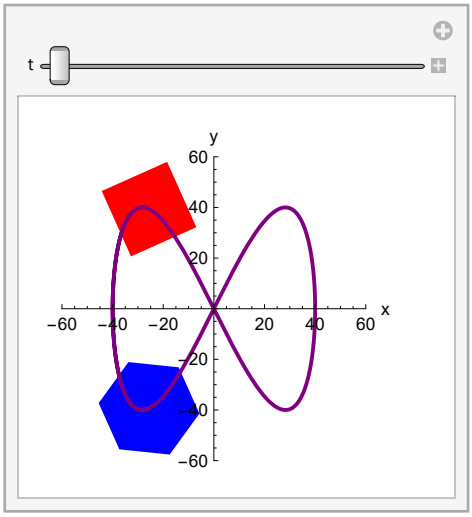
Out[124]=

```
GAll$15493
```

In[125]:=

```
Manipulate[M7[t / 5, t], {t, -20, 20}]
```

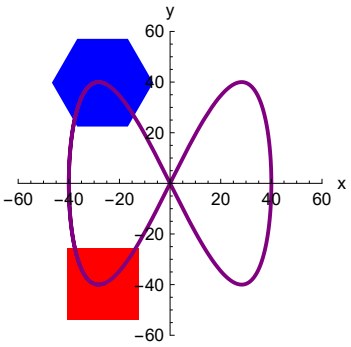
Out[125]=



In[126]:=

```
M7[0, 0]
```

Out[126]=



In[127]:=

```
Rot7[O1_, O2_, OTraj_, Traj_, trange_] :=
 Module[{TAll, MTran1, MTran2, Trans1, Trans2, t1, a1, GAll},
  GAll[t1_, a1_] := Graphics[{GeometricTransformation[O1,
       Composition[TranslationTransform[Traj[-t1 + trange]], RotationTransform[a1]]],
     GeometricTransformation[O2, Composition[TranslationTransform[Traj[t1 - trange]],
       RotationTransform[a1]]], OTraj}, Axes → True, PlotRange → 60];
  GAll]
```

In[128]:=

```
Manipulate[Rot7[{Red, RegularPolygon[{0, 0}, 20, 4]},
    {Blue, RegularPolygon[{0, 0}, 20, 6]}, o62, TrajNew, trange][t / 5, t], {t, -20, 20}]
```

Out[128]=

